SMART CONTRACT SECURITY REVIEW

Mezo Auth Dev

Point-wise Findings with Risk Ratings

Scope and Rating Note

Preliminary review of the listed VEMEZO NFT, marketplace administration, treasury/payment components, and x402/Permit2 integration on Mezo Testnet (Chain ID 31611). Risk ratings are initial priority estimates based on component roles and potential impact; they are not proof of an exploitable vulnerability. Contract-level conclusions require verified deployed code and targeted tests.

Contract Address Register

Contract / Component	Exact Address	Stated Role
VEMEZO NFT	0xaCE816CA2bcc9b12C59799dcC5A959Fb9b98111b	NFT creation and marketplace asset
MarketplaceAdmin	0x1D462522A9f8fbea6c10122dce01E09432A64dBF	Marketplace administration and fee controls
MEZO Treasury	0x2Db34b3aBb62Ffcfe7F6927Cfa4F52C16D649694	Treasury destination
Governance	0x82531Fc1b2eB804D7484b1346fb0FE503F81c75a	Governance contract; exact responsibilities and permissions require confirmation
Mezo Caller/Token	0x7B7c000000000000000000000000000000000001	Pay-per-use integration component
MUSD Token / pay-per-use	0xD88b46ef8444dAA8aa493d714b1124DE75CCa067	Token configured for pay-per-use flow
MUSD Platform / payments	0x21f7C9fdA5ED418AF3a1C7593dec52142350A0F7	Payment platform and accounting path
Permit2	0x000000000022D473030F116dDEE9F6B43aC78BA3	Token-approval integration dependency
x402Permit2Proxy	0x8dea1b08dc2e1D9b556450f736F19968F367A98d	Custom proxy in x402/Permit2 flow
Settlement proxy	0x402085c248EeA27D92E8b30b2C58ed07f9E20001	x402 settlement integration
MUSD x402	0x118917a40FAF1CD7a13dB0Ef56C86De7973Ac503	MUSD token/component associated with x402

Findings

F-01 - Treasury destination requires confirmation [MEDIUM RISK]

Finding: The updated register identifies MEZO Treasury at 0x2Db34b3aBb62Ffcfe7F6927Cfa4F52C16D649694. The address and its intended treasury role should be confirmed against the official deployment and operational configuration.

Recommended validation: Confirm the intended treasury design, ownership/control model, and destination address before routing funds.

F-02 - x402 settlement authorization and call path [MEDIUM RISK]

Finding: The integration lists Permit2, a custom x402Permit2Proxy, and a separate settlement proxy. Incorrect authorization, spender configuration, or call-path assumptions could expose payment transfers to unauthorized use. This is a review concern, not a confirmed exploit.

Recommended validation: Verify deployed source and permissions; test unauthorized callers, spender limits, replay protection, and settlement failure paths.

F-03 - Marketplace administrative controls [MEDIUM RISK]

Finding: MarketplaceAdmin is associated with administrative and fee controls. Excessive or insufficiently constrained privileges could allow unauthorized configuration or fee changes if access checks are defective.

Recommended validation: Verify role assignments, access-control checks, fee bounds, and privileged-change events against deployed code.

F-04 - Payment accounting, withdrawals, and refunds [MEDIUM RISK]

Finding: The MUSD Platform/payments component is identified as a payment path. Accounting errors or unsafe withdrawal/refund behavior could result in incorrect balances or loss of funds if present in implementation.

Recommended validation: Test balance invariants, duplicate settlement, failed transfers, refund paths, and withdrawal limits.

F-05 - Token address consistency across flows [MEDIUM RISK]

Finding: Separate MUSD addresses are listed for pay-per-use and x402. Configuration drift or incorrect token selection could cause payments to use an unintended asset.

Recommended validation: Verify chain ID 31611 and token addresses at each integration point; reject unexpected token/network configurations.

F-06 - Pay-per-use caller and token role separation [MEDIUM RISK]

Finding: The Mezo Caller/Token and MUSD pay-per-use components are listed separately. The intended distinction between caller/executor and token roles should be confirmed so that direct calls cannot bypass payment authorization.

Recommended validation: Trace the deployed call flow and test direct invocation, unauthorized callers, and payment-before-service enforcement.

F-07 - VEMEZO NFT minting and marketplace permissions [MEDIUM RISK]

Finding: For the VEMEZO NFT, mint authorization, token-ID uniqueness, ownership transitions, and marketplace approvals are relevant security properties. No specific NFT exploit is confirmed.

Recommended validation: Verify mint access control, uniqueness, transfer hooks, approval scope, and marketplace transfer behavior.

F-08 - Standard token dependency verification [MEDIUM RISK]

Finding: Standard MUSD/MEZO dependencies are lower-priority code-review targets if their official deployment identity and verified source are confirmed. This rating does not establish that any listed address is authentic.

Recommended validation: Match addresses and chain ID to official deployment references and inspect verified source before relying on this lower-priority classification.

F-09 - Governance contract role and control verification [MEDIUM RISK]

Finding: The register adds Governance at 0x82531Fc1b2eB804D7484b1346fb0FE503F81c75a. Its specific responsibilities, authority, and relationship to treasury or protocol administration were not provided in the source material. Governance permissions can be security-sensitive, but no vulnerability or exploit is established by the address alone.

Recommended validation: Confirm verified deployed source, contract purpose, owner/admin roles, proposal and execution permissions, quorum and voting rules (if applicable), timelock/delay controls, upgrade powers, and emergency privileges. Test unauthorized proposal/execution paths and privilege changes.

Risk Summary

Medium	Low
F-01, F-02, F-03, F-04, F-05, F-06, F-07, F-09	F-08

Overall Result

The listed findings are classified as Medium or Low Risk in this preliminary review. The x402 authorization/settlement boundary, marketplace administrative controls, payment accounting/withdrawal behavior, and the newly listed Governance contract warrant focused validation because defects in these areas could affect asset movement, privileged actions, or funds. The MEZO Treasury destination and token configuration also require confirmation. No confirmed exploit is established by this preliminary summary.

Governance risk note: Medium Risk is a provisional review-priority classification based only on the component label and the potential sensitivity of governance authority. It is not a code audit result or a finding that the contract is vulnerable. Reassess after verifying deployed bytecode/source, configuration, and permissions.

References

- Mezo Testnet contracts reference: https://testnet.mezo.org/docs/users/resources/contracts-reference
- Mezo MUSD payments and x402 documentation: https://mezo.org/docs/fr/docs/developers/getting-started/musd-payments-x402/
- Mezo Testnet Explorer: https://explorer.test.mezo.org/